

CROPS & THEIR CLASSIFICATION

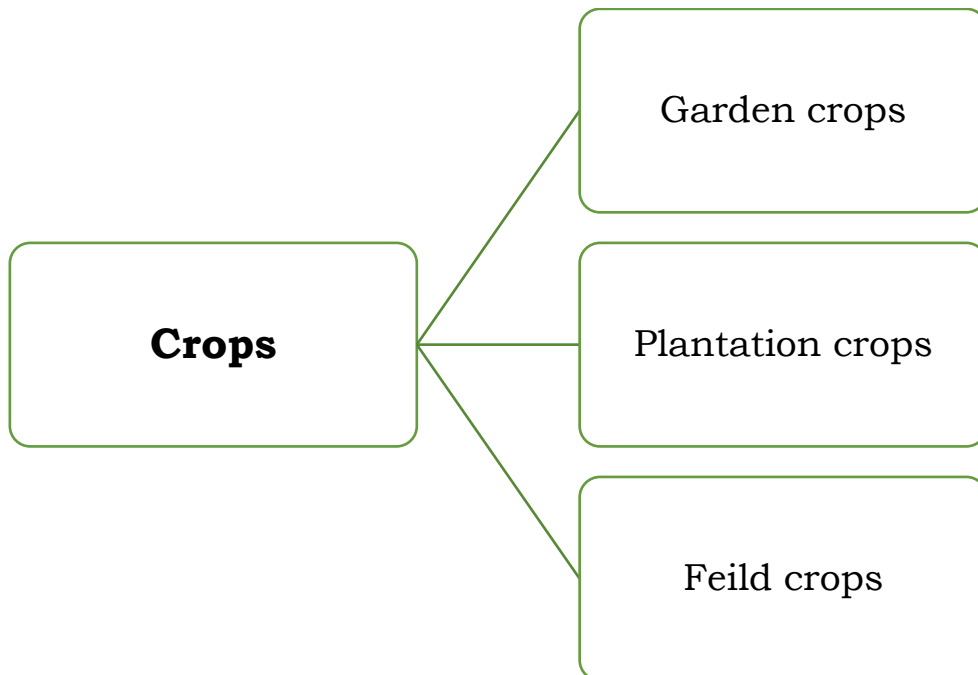
Crop, in agriculture is **a plant or plant product that can be grown** and harvested extensively for profit or subsistence. **Ex -wheat, rice, pulses**

Classification Of Crops

Need of classification-

- ✓ To know about **their habitat, adaptation, life cycle, nutrient requirement**, insect pest & disease reaction of plants.
- ✓ To know about **appropriate production technology**.
- ✓ For **adoption of best management practices to get maximum profit** in minimum input

Crop Plants Are Grouped Into Three Main Classes According To The Range Of Cultivation



Garden crops-

Crop plants that are grown on a small scale in gardens such as kitchen gardens, flower gardens and backyard gardens.

Plantation Crops:

Crop plants that are grown on a larger scale on estates. They are perennial or permanent in nature.

Field Crops:

Crop plants that are grown on a vast scale. They **are mostly seasonal. Among** these three classes agronomy deals with field crops only & horticulture deals with the garden & the plantation crops.

FIELD CROPS

Field crops are **herbaceous plants grown in cultivated fields** under more or less extensive system of culture **ex rice, wheat, pulses, fibre crops etc.**

Basis of classification of field crops: -

1. Botanical aspect
2. Origin
3. Season of growth.
4. Life period.
5. Cultural practices.
6. Agronomical & economical aspects
 - Food crops
 - Non-food crops
 - Special purpose crops
7. Climatic conditions
8. Photoperiod
9. Growth habit
10. Mode of pollination
11. Mode of Reproduction
12. Mode of respiration

1. Classification Of Field Crop On The Basis Of Botanical Aspect:

- a. Monocotyledons
- b. Dicotyledons

1. Monocotyledons-

A plant having only **one seed leaf or cotyledon in each of its seed**. Such plants have a **shallow roots, narrow leaves and tillering** ability.

Example-

- ✓ **Poaceae** - e.g. Cereals i.e. paddy, wheat, sorghum, pearl millet, maize, sugarcane, etc.
- ✓ **Zingiberaceae** - e.g. Ginger
- ✓ **Liliaceae**- e.g. Onion

2. Dicotyledons-

A plant **having two seed leaf or cotyledons in each of its seed**. Such plants have a deep taproots, broad leaves and branching ability.

Examples-

- ✓ **Malvaceae:** e.g. cotton, okra
- ✓ **Cruciferae-** e.g. cabbage, mustard
- ✓ **Solanaceae:** e.g. tobacco, potato, brinjal
- ✓ **Leguminosae:** e.g. groundnut, pigeon pea, gram,
- ✓ **Tiliaceae:** e.g. jute, sun hemp
- ✓ **Linaceae:** eg. linseed
- ✓ **Euphorbiaceae :** e.g. castor
- ✓ **Composite :** e.g. sunflower
- ✓ **Chenopodiaceae:** e.g. sugar beet

2. On The Basis of Origin: -

Native - Crops grown within the geographical limits of their origin, for **e.g., rice, barely, black gram, green gram, mustard, castor, sugarcane and cotton, grown in India**, are native to India.

Exotic or Introduced - Crops introduced from other countries, **such as tobacco, potato, jute, maize, apple, etc.**

3. On The Basis Of Season Of Growth: -

A. Kharif or Monsoon/ Warm season crop:

✚ Crops are grown in the **month of June – July & harvested in the month of September-October**. These crops **require high temperature & onset of monsoon** for sowing while they are harvested at the time of retreating monsoon.

✚ **Optimum temperature** for their growth **is 30 to 38° C** and **maximum temperature** they can **tolerate is 45 to 50 ° C**.

✚ Example- rice, millets, soyabean, cotton, groundnut, sorghum etc

B. Rabi Crops Or Winter Crops/ Cool Season Crops:

✚ Crops are grown **in the month of October – November** and harvested during march to April. These crops **require low temperature during sowing and growth** period but require high temperature during the ripening stage.

✚ **Optimum temperature for rabi crop is 25 to 30° C** & the **minimum temperature ranges between 0 to 5 ° C**

✚ **Example**-Wheat, barley, oats, canola, lentil and mustard.

C. Zaid crop-

Those crops which are **planted in February- March** and harvested in **May- June**.
Example- cucumber, melon, tobacco etc

4. On The Basis Of Life Period

Annuals: A plant which grows from seed, **complete its life cycle and dies during the season** or same year. e.g. cereal crops.

Biennials: A plant which **completes its life cycle in two season or year**. During first season/year, **they complete its vegetative growth and during second season/year** they **completes its reproductive growth**. eg. sugarcane, sugar beet, banana, onion.

Perennial: A plant which completes its life **cycle in more than two seasons**/years. e.g., agave, elephant foot.

5. On The Basis Of Cultural Practices: -

- ✓ **Irrigated crops:** e.g. Sugarcane, paddy, banana etc.
- ✓ **Dry farming crops;** The crop which are grown under natural precipitation. **e.g. sorghum, bajra, groundnut, nagli, etc.**
- ✓ **Sole crops:** One crop variety grown alone in pure stands at normal density.
- ✓ **Monocropping:** The repetitive growing of the same crops on the same piece of land. e.g. **groundnut in Saurashtra region.**
- ✓ **Intercropping:** Growing two or more crops simultaneously on the same piece of land in different rows **e.g., sorghum + black gram, pigeon pea + ground nut, sugarcane + onion, etc.**
- ✓ **Mixed cropping:** Growing two or more crops simultaneously on the same land without/ irrespective to definite row pattern. **e.g. sorghum + green gram+ black gram.**

6. According to Agronomical or Economical Aspects

A Food crops

- ✓ **Cereals:** e.g. paddy, wheat, maize, sorghum, bajra, nagli, etc.
- ✓ **Pulses:** e.g. gram, green gram, black gram, pigeon pea, cowpea, Indian bean, moth bean, etc.
- ✓ **Legumes:** e.g. groundnut, soybean, cluster bean
- ✓ **Edible oil seeds:** e.g. groundnut, sesamum, sunflower, mustard, safflower
- ✓ **Fruit crops:** mango, banana, sapota, guava, papaya, ber, grape citrus crop, etc.

B. Vegetable crops:

- ✓ **Leafy:** cabbage, fenugreek, palak
- ✓ **Fruit:** tomato, okra, brinjal

- ✓ **Root:** radish, carrot
- ✓ **Tuber/Stem:** potato, sweet potato, ginger, turmeric
- ✓ **Bulb:** onion, garlic
- ✓ **Flower:** cauliflower
- ✓ **Pod:** French bean, pigeon pea, cowpea, Indian bean

B) Non-food crops

- ✓ **Forage/fodder** - lucerne, berseem, hybrid nappier grass
- ✓ **Fibre crop:** cotton, jute, sun hemp
- ✓ **Non edible oilseeds:** castor, linseed
- ✓ **Sugar crop:** sugarcane, sugar beet
- ✓ **Dyes:** safflower, indigo
- ✓ **Narcotics:** tobacco, coffee, tea, opium, poppy, chicory
- ✓ **Drugs/ Medicinal:** isabgul, jethimadh, senna, kariyatu
- ✓ **Spices & condiments:** cardamom, cumin, black pepper, coriander, fennel, fenugreek

(C) Special Purpose Crops

- ✓ **Row crops** - Crops which are grown in rows with uniform spacing throughout the field.
e. g. cotton, castor, sorghum, etc.
- ✓ **Support crops**- certain fast-growing crops work as supporter to vine crops. e.g. castor, shevri in betel vine, sorghum in cowpea/bean.
- ✓ **Wind breaks crops**- Crops which are grown on boundaries to protect the field crops from wind. Ex eucalyptus, bamboo etc.
- ✓ **Cover crops**- Cover crops are grown primarily to cover the soil and to reduce the loss of moisture and erosion by wind and water. e.g. groundnut, kidney bean, cowpea, mung bean.

- ✓ **Silage crops:** Crop which are preserved in a succulent condition by partial fermentation in a tight silo pit. e.g. maize, sorghum, bajra.
- ✓ **Cash crops:** crop grown for sale and brings money immediately. **e.g. cotton, tobacco, potato, sugarcane.**
- ✓ **Green manure crops:** Any crop which are grown and buried into the soil for improving the soil condition by the addition of organic **matter. e.g. sun hemp, dhaincha, glyricidia.**
- ✓ **Pasture crops:** Different types of vegetation found on pastures or grassland area which **usually grow. e.g. dharo, zinzvo.**
- ✓ **Catch crops:** Crop which is grown as substitute for the main crop that has failed on account of unfavourable condition. e.g. cowpea, sesame, green gram.
- ✓ **Trap crops:** Crop which is grown on boundary of the field for protection against pest, insect, disease. Marigold, tobacco.
- ✓ **Nurse crops:** Crop which is used to protect or nurse the other crops in their young stage. e.g. cluster bean in ginger, sun hemp in sugarcane.
- ✓ **Companion crops:** Two crops are taken together with the aim that they are benefited to each other. e.g. maize and green gram.
- ✓ **Mixed crops:** Two or more than two crops are grown together on the same piece of land at the same time. e.g. bajra + cowpea + green gram.

7. Based on Climatic Condition

Tropical crop:

Tropical crops are plants that **require a hot and humid climate to grow**. They are typically found in regions that experience little to no frost, and where temperatures average **above 18°C throughout the year** **maize, sorghum coconut, sugarcane**

Sub-tropical crop:

Sub-tropical crops are plants that can tolerate cooler temperatures than tropical crops. They are typically found in regions that experience some **frost, but where temperatures do not drop below -5°C**. Examples of sub-tropical crops include example rice

Temperate crop:

Are those crops which **require low temperature for their growth & development** and can tolerate low **temperature or frost. wheat, barley**

8. On the Basis of Photoperiod:

Crop development like rate **of leaf production, flowering, reproductive stage** etc are **influenced by the duration of sunlight**. The effect of light on plants is called as **photoperiodism** hence depending upon the length of photoperiod (available sunlight) crops are classified in to three categories

a) Long day plants-

These plants require long **duration of sunlight maximum 12 hr sunlight** is required for their vegetative and reproductive stage. If subjected to short day light their vegetative growth may continue but retarded reproductive growth. **They initiate flowering when days are long. Examples – wheat, barley, oats etc.**

b) Short day plant:

Short day plant requires **daylength of less than 10 hours** for their reproductive growth. Example rice, sorghum maize etc.

c) Day neutral plants:

These plants **do not require either long day or short day for their growth & development**. Light periods do not have any impact on their growth phases. Example- cotton, sunflower etc.

9. Based on Growth Habit

1. Determinate plants: Plants which **initiate their reproductive growth after completing** vegetative growth. **Examples: Wheat, barley and rice etc.**

2. Indeterminate plants:

Plants **whose vegetative and reproductive stages continue simultaneously**. Mature and immature fruits, flowers and flower buds are all present on one plant at the same time. **Examples: Tomato, cucumber, watermelon and soybean etc.**

10. Based on Mode of Pollination

Pollination: Transfer of pollen from anther to stigma of flowers.

- ✓ **Self-pollinated plants/ autogamy**- Transfer of pollen from anther of same flower to stigma of same flower of same plant. **Usually 1-3% cross** pollination occurs in these plants. **Examples: Rice, wheat, green gram, black gram and soybean etc**
- ✓ **Cross pollinated plants/ allogamy** - Transfer of pollen from anther of one flower to stigma of another flower of same plant or different plant. **Examples: Maize, bajra, sunflower and brassica etc.**

11. Based on Mode of Propagation:

- ✓ **Sexually Propagated Plants:** Plants are propagated through seeds are sexually propagated plants. **Examples: Rice, wheat, green gram, black gram, soybean, maize and barley etc.**
- ✓ **Asexually Propagated Plants:** Plants **are propagated asexually by using specialized parts** or utilizing cuttings, grafting and layering etc. **Examples: Sugarcane, ginger and potato etc**

12. Based on Mode of Respiration:

1. C3 plants:

- ✓ Plants fix CO₂ and form a 3- carbon molecule **and 3- phosphoglyceric acid (3 PGA)**.
- ✓ This pathway was discovered **by Calvin and his co-workers** (Bassham and Calvin, 1957).
- ✓ Plants with this pathway of **carbon assimilation are called C3 pathway plants**. ***These plants do exhibit photo-respiration.***

- ✓ **Such plants cannot utilize light, temperature** water and CO₂ effectively, therefore they are called as Inefficient plants.
- ✓ **Examples:** wheat, oat, rye, soybean and cotton etc.

2. C₄ plants:

- ✓ In these plants, first product of **photosynthesis is a 4- Carbon molecule called Oxalo** acetate.
- ✓ Plants fix **CO₂ via Hatch and Slack pathway**. These plants do not exhibit photorespiration.
- ✓ Such plants can utilize **light, temperature water and CO₂** effectively, therefore they are called as Efficient plants. **Examples: Sugarcane, maize and sorghum etc.**

3. CAM plants (Crassulacean acid metabolism) pathway:

- ✓ **CAM Plants fix CO₂ into 4- carbon acids** called malate as do the C₄ plants. Fixation of occurs at night when stomata open.
- ✓ CAM plants **grow in deserts and have succulent fleshy leaves** and stem with low transpiration and water requirement. **Examples: Pine apple and Preckly pear etc**